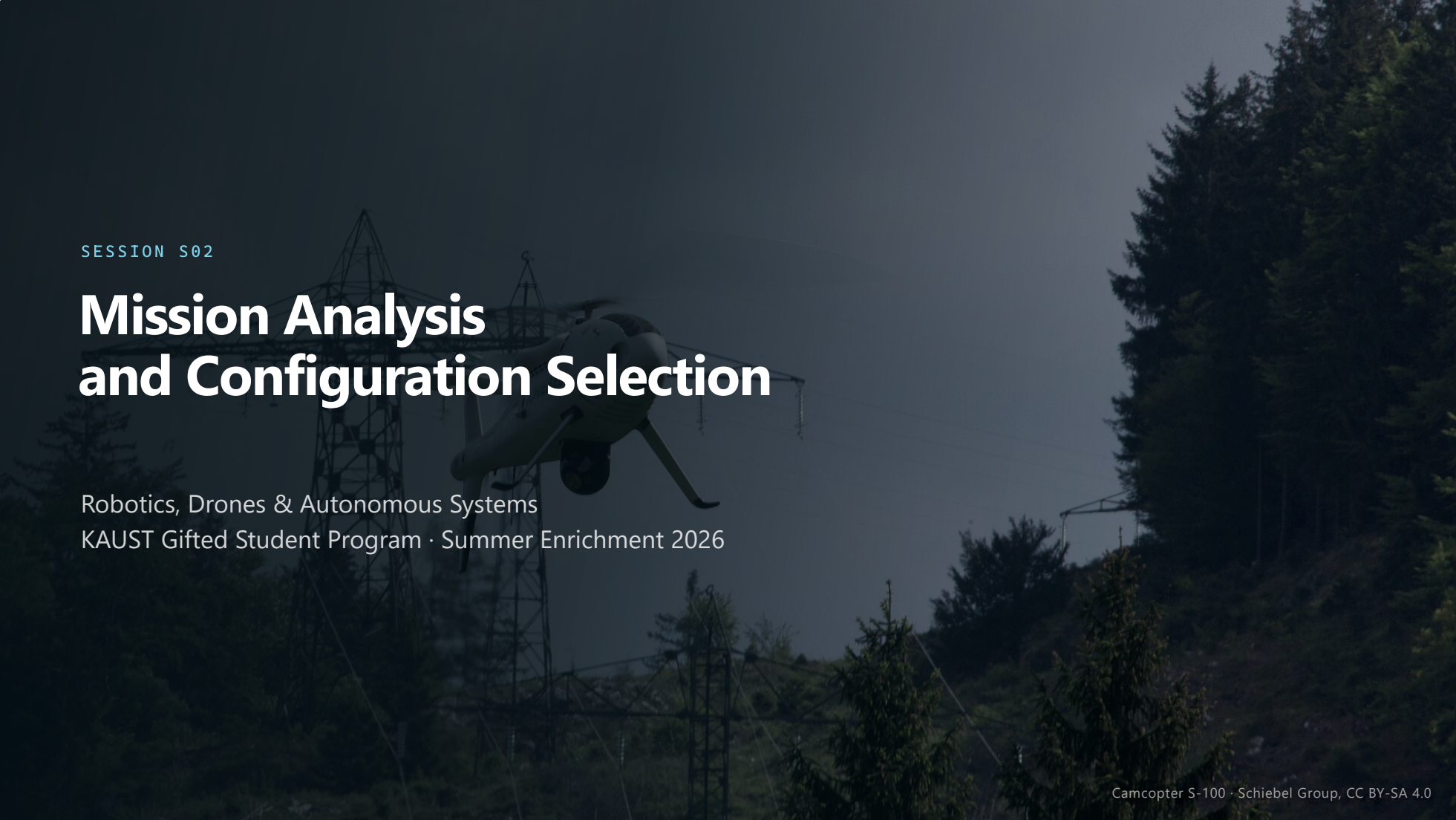


The background image is a dark, atmospheric photograph of a forested hillside. A tall, lattice-structured radio tower stands prominently on the left side. A drone, identified as a Camcopter S-100, is captured in flight in the center-left area, its rotors blurred. The sky is a deep, overcast blue-grey. The overall mood is technical and serene.

SESSION S02

Mission Analysis and Configuration Selection

Robotics, Drones & Autonomous Systems

KAUST Gifted Student Program · Summer Enrichment 2026

WHERE WE ARE

**By the end of this session
you will have selected a configuration.**

Session outline

- **Your problem, the numbers behind it, then how to decide**

Instructor-led. The only lecture in this session.

- **Three requirements, then screen and score**

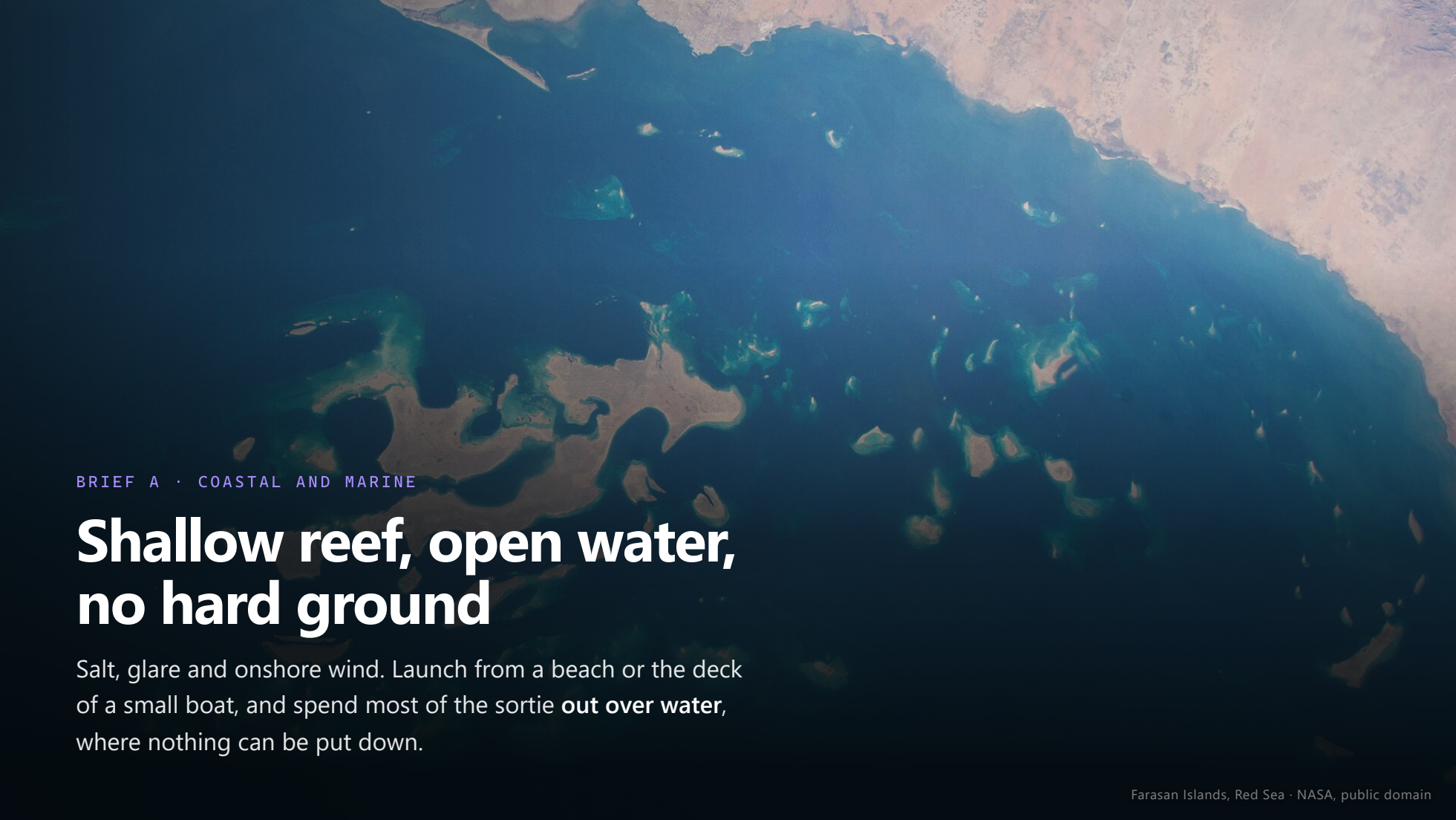
One exercise in two halves. The second half needs the first.

- **Review**

Not what you chose — why.

This session is a **complete project on its own**. It begins with a problem statement and ends with a chosen configuration. S03 is a different project.

Output: Report 1 — your configuration choice, **and every option you rejected, with the reason**. Worth 15%. The rejections are marked as heavily as the choice.



BRIEF A · COASTAL AND MARINE

Shallow reef, open water, no hard ground

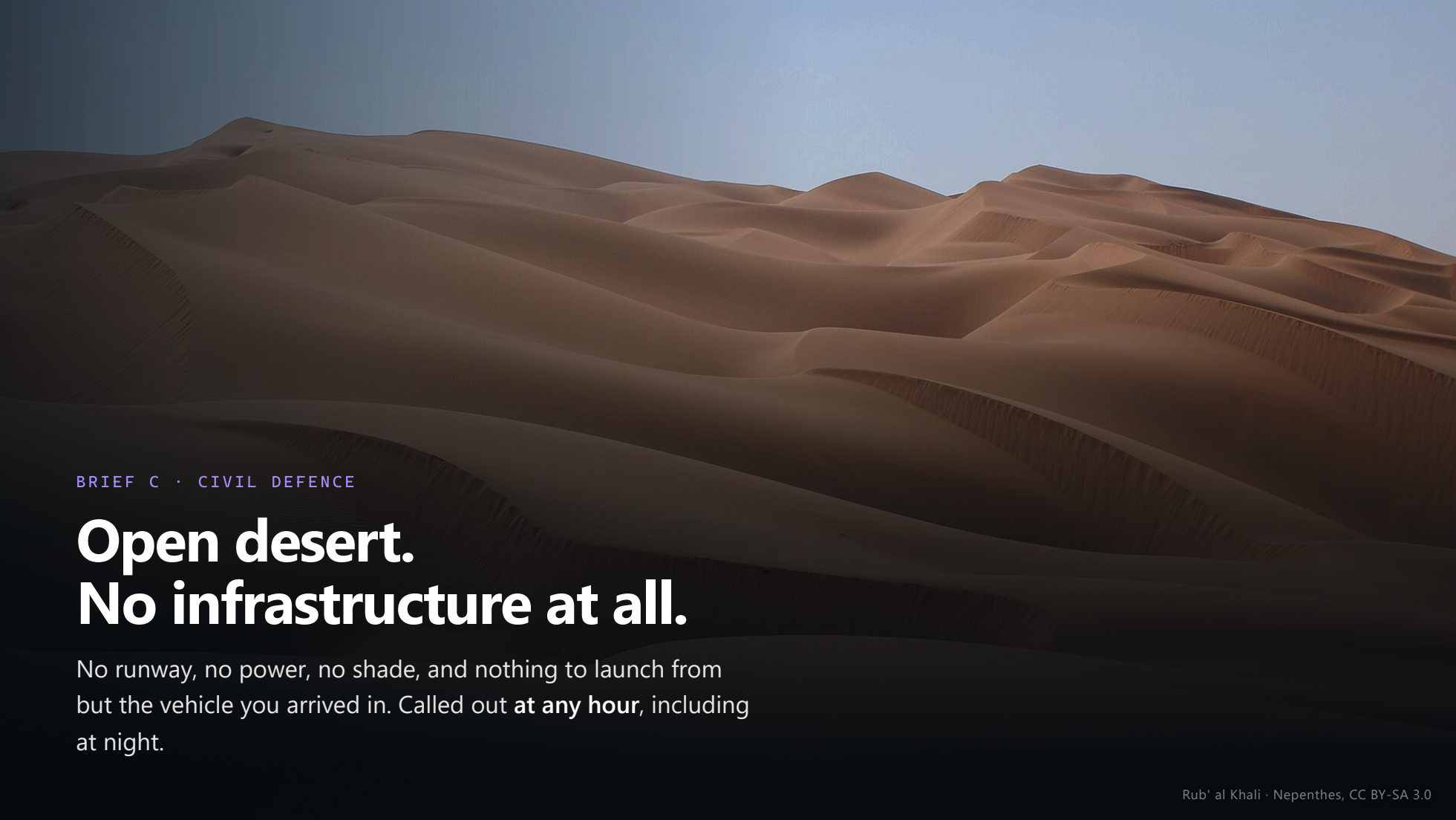
Salt, glare and onshore wind. Launch from a beach or the deck of a small boat, and spend most of the sortie **out over water**, where nothing can be put down.



BRIEF B · PLANT INTEGRITY

Steel, concrete, and a plant that will not stop

Structure in every direction, hot plumes above the process equipment, and **GPS you cannot trust** between the pipe racks. The work happens while the plant runs.



BRIEF C · CIVIL DEFENCE

Open desert. No infrastructure at all.

No runway, no power, no shade, and nothing to launch from but the vehicle you arrived in. Called out **at any hour**, including at night.

Three problems, three sets of numbers

Three environments, three problems, three sets of numbers. **Nobody here is solving the same problem as the team beside them** — and your requirements should show it.

If your three requirements would work just as well for somebody else's brief, they are **too vague to be requirements**. That is the fastest test you can run on your own list in this session — and it is the one I will run at the review.

You will hear the other two briefs at the end of the session. Until then, work yours.

PART ONE

Defining requirements

A requirement states what, not how

NOT A REQUIREMENT

A REQUIREMENT

Left names a feeling or a part; right names a number someone can check. Every one of yours needs a **subject**, the word **shall**, a **quantity with units** and a **way to verify it**.

The vehicle shall maintain position within ± 2 m of each waypoint in winds up to 6 m/s, verified from the flight log.

A requirement states what, not how

NOT A REQUIREMENT

- "The drone should have good battery life."

A REQUIREMENT

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NOT A REQUIREMENT

- "The drone should have good battery life."
- "Use a multirotor with six motors."

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NOT A REQUIREMENT

- "The drone should have good battery life."
- "Use a multirotor with six motors."
- "Fly the survey quickly."

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The vehicle **shall** maintain position within ± 2 m of each waypoint in winds up to 6 m/s, verified from the flight log.

A requirement states what, not how

NOT A REQUIREMENT

- "The drone should have good battery life."
- "Use a multirotor with six motors."
- "Fly the survey quickly."

A REQUIREMENT

- "Complete the 600 m survey, land with $\geq 20\%$ battery."

Left names a feeling or a part; right names a number someone can check. Every one of yours needs a **subject**, the word **shall**, a **quantity with units** and a **way to verify it**.

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A REQUIREMENT

- "Complete the 600 m survey, land with $\geq 20\%$ battery."
- "Remain controllable after any single motor failure."

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NOT A REQUIREMENT

- "The drone should have good battery life."
- "Use a multirotor with six motors."
- "Fly the survey quickly."

A REQUIREMENT

- "Complete the 600 m survey, land with $\geq 20\%$ battery."
- "Remain controllable after any single motor failure."
- "Cover 500×400 m at ≤ 3 cm/pixel in one sortie."

Left names a feeling or a part; right names a number someone can check. Every one of yours needs a **subject**, the word **shall**, a **quantity with units** and a **way to verify it**.

The vehicle shall maintain position within ± 2 m of each waypoint in winds up to 6 m/s, verified from the flight log.

Requirements and constraints are not the same

REQUIREMENT

Something your design must **achieve**.

You chose it, from the mission. You can argue for a different number if you can justify it.

"Shall cover the full survey area on one battery."

Requirements score. They become the rows of your matrix.

CONSTRAINT

Something your design must **live inside**.

You did not choose it and you cannot argue with it. Budget, physics, regulation, the site you fly from.

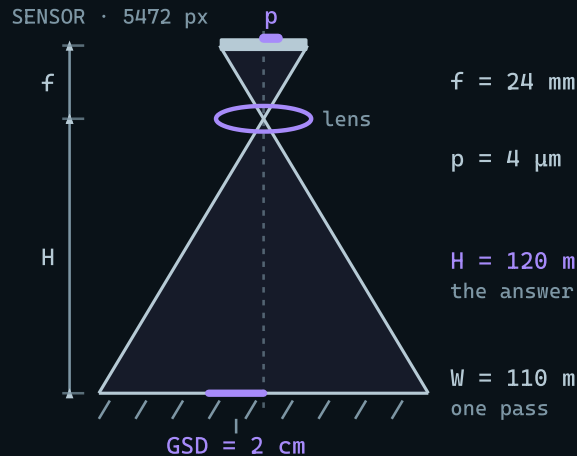
"Shall operate at 45 °C without thermal shutdown."

Constraints filter. They throw options out before scoring starts.

Mixing these up costs you twice. Score an option that a constraint should have killed, and you will spend twenty minutes carefully ranking something you cannot build. **Sheet 2 lists the seven places a constraint hides** — budget, payload, crew, launch and recovery, environment, schedule, transport. Work it row by row.

From qualitative statement to quantified requirement

The problem states "2 cm per pixel". That is not yet a requirement — it is **one side of a triangle**.



Two triangles, one apex. $GSD/p = H/f$ — so the resolution fixes the height.

GROUND SAMPLE DISTANCE · WHAT THE CAMERA CAN RESOLVE

$$GSD = \frac{p \cdot H}{f} \implies H = \frac{GSD \cdot f}{p}$$

the required resolution has fixed the altitude you may fly at

GSD Ground distance one pixel covers — **the number in your brief**

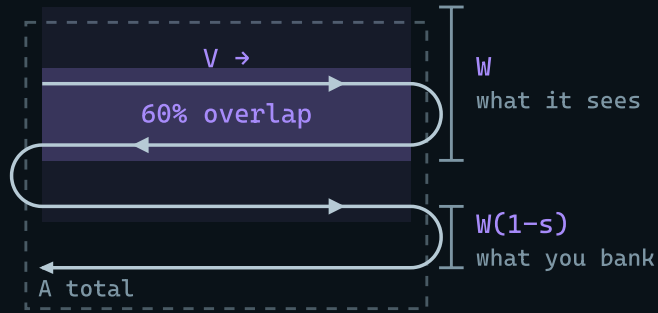
p Pixel pitch on the sensor, typically 2–4 μm

f Lens focal length, mm

H Height above the ground — **the answer**, not a choice

Fly higher, cover more per pass, break the resolution. Fly lower, hold it, fly more passes. **The resolution requirement has already fixed your altitude** — it was simply not stated that way.

From area and schedule to required endurance



The two shaded passes photograph the band between them **twice**. That is what the overlap costs you.

COVERAGE RATE · WHAT ONE AIRCRAFT CAN ACTUALLY SWEEP

$$\dot{A} = W(1-s)V \quad t_{\text{required}} = \frac{A_{\text{total}}}{\dot{A}}$$

$\dot{A}$ Coverage rate — ground area swept per second, m^2/s

W Swath the camera sees across track. $W = \text{GSD} \times \text{pixels across}$

s Side overlap required — **60% means you bank only 40%**

V Ground speed — **yours to choose**, the only free variable

A_{total} The whole area to be surveyed, m^2

t_{required} Flying time it demands — **the answer**

WHAT THE BRIEF SAYS

"Survey **4.8 km²** at **2 cm/pixel**, **60% overlap**, in one working day."

Three numbers and an adjective. Not testable.

WHAT IT BECOMES

A **20 MP** sensor sees **110 m**; you bank **44 m** a pass.

At **12 m/s** → **1.9 km²/h** → **about 2.5 flying hours**.

One wing sortie. **Six** multirotor battery swaps.

PART TWO

The selection method

Two steps, and the order matters

01

Screen against your constraints

Binary. Pass or fail. An option that violates a constraint is *out* — it does not get a score, it gets one line saying which constraint killed it.

02

Score the survivors against your requirements

Weighted. Everything that got through screening is genuinely viable, so now you are asking which is *best*, not which is *possible*.

Constraints filter. Requirements score. That is the whole reason we separated them ten minutes ago — and it is why screening comes first.

Step one: screening

Worked example. **Not your brief** — a property firm wants 20 house exteriors photographed a day: one operator working out of a car, 8 images of each from 40 m, \$9,000 a system.

Fixed-wing	OUT	Nothing to launch from and nothing to land on in a suburban street
------------	-----	--

Helicopter	OUT	\$9,000 does not buy one, let alone maintain one
------------	-----	--

Multirotor	IN	Carried forward to scoring
------------	----	----------------------------

VTOL hybrid	IN	Carried forward — the wing is a weak fit, not a violated constraint
-------------	----	---

Half the options are gone and not one number has been scored yet. **Screening is where most of the work happens** — and those two rejection lines are already report content.

Step two: weighted scoring

CRITERION	FROM	WEIGHT	MULTIROTOR	VTOL HYBRID
≤ 6 minutes per property	R1	5	4	5
Set up in ≤ 2 min from the car	R2	4	4	3
Spares available locally	R3	2	5	3
Weighted total			46	43

Criteria are your **three requirements**, one per row. Weights come from your own ranking — most important becomes 5. Scores are 1–5 against the criterion, judged by you and **defended by you**.

46 to 43 is **not a decision**, it is a tie. Three points out of forty-six is smaller than the error in scores you assigned by hand. When that happens: find the criterion you left out, or challenge a weight from your brief — and say which you did.

A QUESTION FOR THE ROOM

Have you already decided what your team is going to build?

Most teams have, and there is nothing wrong with that. The discipline is scoring the matrix honestly regardless — and being willing to be surprised by your own numbers.

60 SECONDS Answer for yourselves. No follow-up questions

Three common failure modes

You decided first and scored backwards. Everyone does it and nobody notices.

→ Fill every score before you total anything. Have one team member score a column without seeing the others.

Every weight is a 3. If it all matters equally, the matrix is doing no work.

→ Force a spread. You ranked your three requirements — use the ranking.

You never wrote down what you rejected. The report shows one option and one winner.

→ The rejections and their reasons are the evidence you did this at all.

PART THREE

Exercise — 40 minutes

Task

Sheets 1–2 are roughly 16 minutes, then move on ready or not. Sheet 3 is roughly 24, and it is the graded half. Screen all four classes, even the ones you know are wrong — a sheet with two options on it is a decision you had already made before you sat down.

CORE · EVERYONE FINISHES THIS

STRETCH · IF YOU FINISH EARLY

Task

Sheets 1–2 are roughly 16 minutes, then move on ready or not. Sheet 3 is roughly 24, and it is the graded half. Screen all four classes, even the ones you know are wrong — a sheet with two options on it is a decision you had already made before you sat down.

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- Read your brief again, properly, as a team

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- Write **three** requirements. Subject, *shall*, a number, a verification method. Rank them

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- Fill the constraints table, and say what each one **rules out**

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- Screen **all four** configuration classes — name the constraint that kills each rejection

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- Score the survivors, total it, and write **one paragraph**: what you chose, and what would change your mind

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- Re-run with one weight changed. If the answer flips, you have found the weight your whole choice rests on — **name it**

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STRETCH · IF YOU FINISH EARLY

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- Was "**one aircraft**" in your brief, or only in your head?

Review — two and a half minutes per team

Tell the room:

- 1 Your mission in one sentence, and what you chose
- 2 The option that was hardest to reject, and the constraint that killed it
- 3 The weight your answer is most sensitive to
- 4 What evidence would change your mind

You are not defending a *right answer*. You are showing that the answer follows from the brief — and **three different briefs should not produce three identical reports.**

NEXT SESSION

S03 — Multirotors

S03 is a **different project**, with different briefs and its own report. The class is decided for you: it is a multirotor. The questions are how many rotors, in what layout, at what size — and then what it is made of.

Here you chose a **class of aircraft**.

Next you design **one particular machine**.

Whatever you chose here, you keep. Report 1 stands on its own.

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